

SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS AND SCIENCE

BCA – Semester 6

Subject: Data Analysis with Python

Chapter 1: Exploratory Data Analysis (EDA)

Practical Assignment - 1

Instructions

1. Use **Python IDLE** tool for all questions.
2. Show **all steps, tables, and graphs** clearly.
3. Write final **observations in 1–2 lines**.

Q1. Univariate EDA – Marks (Spread)

Marks of 10 students are:

40, 45, 50, 55, 60, 65, 70, 75, 80, 85

Tasks:

- a) Calculate: mean, median, mode, minimum, maximum, and range.
- b) Draw a histogram of marks.
- c) Comment in one line on the spread of data.

Q2. Univariate EDA – Skewed Marks

Marks of 10 students are:

10, 15, 20, 25, 30, 35, 40, 80, 85, 90

Tasks:

- a) Calculate mean and median.
- b) Draw a histogram of marks.
- c) State whether the distribution is left-skewed, right-skewed or approximately symmetric.

Q3. Categorical Data – Favourite Programming Language

Favourite programming language of 12 students:

C, Python, Java, C, Python, Python, C++, Java, C, Python, C, Java

Tasks:

- a) Prepare a frequency table for each language.
- b) Draw a bar chart for language vs frequency.
- c) Identify the most popular language.

Q4. Bivariate Analysis – Hours Studied vs Exam Score

Dataset of 10 students:

Student	Hours_Studied	Exam_Score
1	1	30
2	2	40
3	2	42
4	3	50
5	3	55
6	4	60
7	4	65
8	5	72
9	6	80
10	7	88

Tasks:

- Draw a scatter plot (Hours_Studied on X-axis, Exam_Score on Y-axis).
- Calculate the correlation coefficient between Hours_Studied and Exam_Score.
- State whether the relationship is positive, negative or no clear relation.

Q5. Multivariate Analysis – Marks in Three Subjects

Dataset of 8 students:

Student	Maths	Science	English
A	60	65	70
B	55	60	58
C	70	75	72
D	80	78	82
E	65	68	66
F	50	52	55
G	75	80	78
H	68	70	72

Tasks:

- Compute the correlation matrix for Maths, Science and English.
- Represent the correlation matrix using a heatmap (if tool supports).
- Identify any one pair of subjects with strong positive correlation.

Q6. Handling Missing Data – Student Marks

Dataset:

Student	Maths	Science
A	70	80
B	60	NaN
C	NaN	75
D	90	85

E	50	NaN
---	----	-----

(Here NaN indicates missing value.)

Tasks:

- Count the number of missing values in each column.
- Replace missing values with the mean of that column (mean imputation).
- Show the table before and after imputation.

Q7. Outliers – Monthly Income

Monthly income (₹) of 12 people:

25,000; 27,000; 28,000; 30,000; 29,000; 26,000; 27,500; 28,500; 26,500; 27,200; 2,00,000; 2,20,000

Tasks:

- Draw a boxplot of the income data.
- Identify the outliers from the boxplot.
- Remove the outliers and recalculate the mean income.
- Write one line comparing the mean before and after removing outliers.

Q8. Data Types – Mixed Dataset

Dataset of 8 entries:

ID	Age	Gender	Height_cm	Rating (1–5)
1	18	F	155	4
2	20	M	170	3
3	19	F	160	5
4	21	M	175	2
5	22	F	162	4
6	23	M	178	3
7	24	F	168	5
8	25	M	180	1

Tasks:

- Classify each column as: Discrete, Continuous, Nominal or Ordinal.
- Prepare a separate table with columns: Variable | Type.

Q9. Skewness – Income Example

Income (₹ in thousands) of 10 people:

10, 12, 13, 14, 15, 16, 18, 20, 50, 60

Tasks:

- a) Calculate mean and median.
- b) Draw a histogram of the data.
- c) State whether the distribution is left-skewed, right-skewed or approximately symmetric.

Q10. Kurtosis – Comparison of Two Classes

Marks (out of 50):

- Class A: 25, 26, 27, 28, 29, 30, 31, 29, 30, 28
- Class B: 10, 15, 20, 25, 30, 35, 40, 45, 20, 38

Tasks:

- a) Draw separate histograms for Class A and Class B.
- b) Compute kurtosis for both Class A and Class B.
- c) Identify which class has a sharper peak (higher kurtosis) and write one-line justification.

Date of Submission: _____

Signature of Teacher: _____